Probability and Computing:
Exercise Sheet 6

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Question 1

Let (X_t, Y_t) be a coupling of the Markov chain M_t. We choose a coupling as follows.

- Choose positions j_1, j_2 u.a.r. from $\{1, ..., n\}$.
- Obtain X_{t+1} from X_t by swapping the card at position j_1 with card at position j_2. Call the card at position j_2 card C.
- Obtain Y_{t+1} from Y_t by swapping the card at position j_1 with card C.

We show that Y_t is a faithful copy of the original chain. Consider any position k_1, k_2. Let C' be the card in position k_2 in Y_t. Let j_2 be the position of C' in X_t, and let $j_1 = k_1$. The probability that j_1, j_2 are chosen are each $1/n$, so this is the probability that position k_1, k_2 are chosen in the transition from Y_t to Y_{t+1}.

Let Z_t denote the random variable of the number of cards whose positions are different for X_t and Y_t. Clearly $0 \leq Z_t \leq n$. At each time step t, we choose a (card, position) pair (C_i, j_i). If C_i in the same position for both, or if the card at j_i is the same for both X_t and Y_t, then $Z_{t+1} = Z_t$. Otherwise, we have $Z_{t+1} \leq Z_t - 1$. The probability of the chosen card C_i being in different positions is Z_t/n, and the probability of the chosen position j_i having different cards is also Z_t/n. Since both events are independent,

$$\Pr(Z_{t+1} \leq Z_t - 1) = \left(\frac{Z_t}{n}\right)^2$$

Let T_i be the time spent for which $Z_t = i$. Then the maximum total time spent before coupling is $T \leq \sum_{i=1}^{n} T_i$, then $\mathbb{E}[T] \leq \sum_{i=1}^{n} \mathbb{E}[T_i]$. T_i is a geometric r.v. with parameter $p_i = \frac{i^2}{n^2}$, so $\mathbb{E}[T_i] = \frac{n^2}{i^2}$. Then, the expected time spent is

$$\mathbb{E}[T] \leq n^2 \sum_{i=1}^{n} \frac{1}{i^2} < \frac{n^2\pi^2}{6}$$

By Markov's inequality, the probability not to obtain the desired sequence after $\frac{n^2\pi^2}{3}$ steps is at most $1/2$. Thus, after $\frac{kn^2\pi^2}{3}$ steps, the probability not to obtain the sequence is at most 2^{-k}. We deduce that $\Pr(X_t \neq Y_t) \leq \varepsilon$ if $t \geq \frac{\log_2(1/\varepsilon)n^2\pi^2}{3}$, so by the Coupling Lemma, the mixing time is

$$\tau(\varepsilon) = O(n^2 \log(1/\varepsilon))$$

Question 2

The Markov chain is **finite**, with state space $\Omega = \mathcal{I}(G) \subseteq 2^V$. It is **irreducible**, since we can get from any independent sets $I \in \mathcal{I}(G)$ to $I' \in \mathcal{I}(G)$ by adding or removing vertices approriately. It is **aperiodic**, since it can self-loop. Thus, there is a unique stationary distribution π.

Let $n = |V|$. Observe that for $I, J \in \mathcal{I}(G), I \neq J$,

- $P_{I,J} = \frac{\lambda}{2n(1+\lambda)}$, if $J \in \mathcal{I}(G)$ has 1 more vertex than I
- $P_{I,J} = \frac{1}{2n(1+\lambda)}$, if $J \in \mathcal{I}(G)$ has 1 less vertex than I
- $P_{I,J} = 0$, otherwise

Consider the distribution of the Gibbs measure, so $\pi_I = \lambda^{|I|}/Z(G)$. Consider also arbitrary $I, J \in \mathcal{I}(G)$, where $|I| + 1 = |J|$.

$$\begin{aligned}
\pi_I P_{I,J} &= \frac{\lambda^{|I|}}{Z(G)} \cdot \frac{\lambda}{2n(1+\lambda)} \\
&= \frac{\lambda^{|I|+1}}{Z(G)} \cdot \frac{1}{2n(1+\lambda)} \\
&= \frac{\lambda^{|J|}}{Z(G)} \cdot \frac{1}{2n(1+\lambda)} \\
&= \pi_J P_{J,I}
\end{aligned} \tag{1}$$

Clearly if $P_{I,J} = P_{J,I} = 0, \pi_I P_{I,J} = 0 = \pi_J P_{J,I}$. Thus, $\forall I, J, \pi_I P_{I,J} = \pi_J P_{J,I}$. By time-reversibility in Theorem 4.2.12, the Gibbs measure is the stationary distribution π.

Suppose the state space Ω is instead the set of all subgraphs of G. The chain still has the same unique distribution π being the Gibbs measure on the independent sets. If a state $I \notin \mathcal{I}(G)$, then the algorithm will have the state only remain the same or remove vertices until it is an independent set, after which, it will not return to being a non-independent set.

Now let H be the graph on Ω in which there is an edge from x to y if and only if x and y differ by only 1 vertex. For all edges $(x, y), d(x, y) = 1$. Since Ω contains all subgraphs, the corresponding distance metric d is just number of vertices that are in either X_t or Y_t, and not both. For all states $(x, y), d(x, y)$ is just $|x - y| + |y - x|$. We choose the coupling where both $(X_t), (Y_t)$ make the same decision in the original Markov chain.

Take $t = 0$ and suppose that $(X_0, Y_0) = (x, y)$ where (x, y) is an edge of H so x and y differ by only 1 vertex. WLOG, suppose $x = I, y = I \cup \{u\}$. Let $d_t = d(X_t, Y_t)$, and let $d_0 = 1$. If $X_1 \leftarrow X_0$, so $Y_1 \leftarrow Y_0$, then $d_1 = d_0 = 1$. Otherwise, consider when the vertex v is chosen u.a.r. from V.

- If $v \in X_0$, so $v \in Y_0$, then $d_1 - d_0 = 0$ unchanged for any decision made.
- Else If $v = u$, then $d_1 - d_0 = -1$ decrease for any decision made.
- Else If $v \in N(u)$,
  - Removing v will leave $d_1 - d_0 = 0$ unchanged.
  - Adding v will have $d_1 - d_0 = 1$ increase, since it can only be added to X_0, and not Y_0.
- Else If $v \notin N(u)$, then $d_1 - d_0 = 0$ unchanged for any decision made.

Analysing all cases, only 2 of them will cause $d_1 \neq d_0$, and noting $d_0 = 1$, so

$$\begin{aligned}
\mathbb{E}[d_1] &= 1 - \frac{1}{2n} + \frac{\lambda}{1+\lambda} \cdot \frac{d(u)}{2n} \\
&\leq 1 - \frac{1}{2n} + \frac{\lambda}{1+\lambda} \cdot \frac{3}{2n} \\
&= 1 + \frac{1}{2n}\left(-1 + \frac{3\lambda}{1+\lambda}\right) \\
&= 1 + \frac{1}{2n} \cdot \frac{2\lambda - 1}{1+\lambda} \\
&= d_0\left(1 + \frac{1}{2n} \cdot \frac{2\lambda - 1}{1+\lambda}\right) \\
&< d_0
\end{aligned} \tag{2}$$

Note that since $0 < \lambda < \frac{1}{2}, -1 < \frac{2\lambda-1}{1+\lambda} < 0$.

Now we can apply the path coupling lemma with $\beta = \left(1 + \frac{1}{2n} \cdot \frac{2\lambda-1}{1+\lambda}\right)$ and $D = n$, since the maximum the maximum number of vertices X_t, Y_t can differ by is n. This shows that the mixing time satisfies $\tau(\varepsilon) = O(\ln(n/\varepsilon)/\ln(1/\beta))$. Since $1 + x \geq e^x$, and $1 + \lambda = \Theta(1)$,

$$\beta \leq e^{\frac{1}{2n} \cdot \frac{2\lambda-1}{1+\lambda}}$$

$$\frac{1}{\beta} \geq e^{-\frac{1}{2n} \cdot \frac{2\lambda-1}{1+\lambda}}$$

$$\ln\frac{1}{\beta} \geq -\frac{1}{2n} \cdot \frac{2\lambda-1}{1+\lambda}$$

$$\frac{\ln(\frac{n}{\varepsilon})}{\ln\frac{1}{\beta}} \leq \frac{\ln(\frac{n}{\varepsilon})}{\frac{1-2\lambda}{2n(1+\lambda)}} = \frac{2n(1+\lambda)\ln(\frac{n}{\varepsilon})}{1-2\lambda}$$

$$\tau(\varepsilon) = O\left(\frac{n\ln(\frac{n}{\varepsilon})}{1-2\lambda}\right)$$

Question 3

Let X_t denote the random variable where $X_t = 1$ and $X_t = -1$ if the clockwise and anticlockwise are taken at step $t \geq 1$, and $X_0 = 0$. Then $Z_t = \sum_{i=0}^{t} X_t$ is the position of the fox at step t.

We treat going anticlockwise from 0 as going into the negative values to the minimum of $-(n-1)$. Clearly we have $|Z_t| \leq n-1$. Then the event that the fox eats sheep i last is the union of the events that it reaches the sheep going clockwise, or anticlockwise, which are clearly

Consider the fox eats the sheep clockwise. Then it must have reached sheep $i+1$ first before reaching i, followed by reaching i from $i-1$ before reaching i from $i+1$. Then this is the same reaching $i+1-n$ before i in random walk on a line in $[i+1-n, i]$, followed by reaching $n-1$ before -1 in a random walk on a line in $[-1, n-1]$

Similarly, if the fox eats the sheep anticlockwise, it reached the sheep $i-1$ first before reaching i, followed by reaching i from $i+1$ before reaching i from $i-1$. This is the same as reaching $i-1$ before $i-n$ in random walk on a line in $[i-n, i-1]$, followed by reaching $1-n$ before 1 in a random walk on a line in $[1-n, 1]$.

Both these events are then consecutive instances of the gambler's fortune, with $a_{1,1} = i, b_{1,1} = n-i-1, a_{1,2} = n-1, b_{1,2} = 1$ in the clockwise case, and $a_{2,1} = i-1, b_{2,1} = n-i, a_{2,2} = 1, b_{2,2} = n-1$. Note that for all cases $a_{i,1} + b_{i,1} = n-1$, and $a_{i,2} + b_{i,2} = n$.

Since we the know the probability of reaching a before b is $\frac{b}{a+b}$, and b before a is $\frac{a}{a+b}$, the probability the fox eats sheep i last is

$$\frac{a_{1,1}b_{1,2}}{n(n-1)} + \frac{b_{2,1}a_{2,2}}{n(n-1)} = \frac{i+n-i}{n(n-1)} = \frac{1}{n-1}$$

This is independent of i, so every sheep is equally likely to be eaten last.

Question 4

Note that

$$Z_j - Z_{j-1} = \sum_{i=1}^{j} X_i - \sum_{i=1}^{j-1} X_i = X_j$$

Then

$$Y_i = \sum_{j=1}^{i} \sum_{k=1}^{j-1} X_k X_j = \sum_{j=1}^{i} X_j \sum_{k=1}^{j-1} X_k$$

so Y_n is a function of $X_0, ..., X_n$, and

$$\begin{aligned} \mathbb{E}[|Y_n|] &= \mathbb{E}\left[\left|\sum_{i=1}^{n} X_i Z_{i-1}\right|\right] \\ &= \mathbb{E}\left[\left|\sum_{i=1}^{n} Z_{i-1}\right|\right] \quad \text{(since } X_i \in \{-1, 1\}\text{)} \\ &\leq \sum_{i=1}^{n} \mathbb{E}[|Z_{i-1}|] \\ &\leq \sum_{i=1}^{n} (i-1) \\ &< \infty \end{aligned} \tag{3}$$

We prove by induction that

$$\mathbb{E}[Y_{n+1} | X_1, ..., X_n] = Y_n$$

Base case: $n = 0$

$$\begin{aligned} \mathbb{E}[Y_2 | X_1] &= \mathbb{E}[X_1 Z_0 + X_2 Z_1 | X_1] \\ &= \mathbb{E}[X_1 Z_0 | X_1] + \mathbb{E}[X_2 Z_1 | X_1] \quad \text{(linearity of expectation)} \\ &= 0 + \mathbb{E}[X_2 X_1 | X_1] \quad \text{(since } Z_0 = 0\text{)} \\ &= \mathbb{E}[X_2] X_1 \quad \text{(since } X_1, X_2 \text{ independent)} \\ &= 0 \quad \text{(since } \mathbb{E}[X_i] = 0\text{)} \\ &= Y_1 \end{aligned} \tag{4}$$

Suppose

$$\mathbb{E}[Y_n | X_1, ..., X_{n-1}] = Y_{n-1}$$

Note that

$$Y_{n+1} = Y_n + X_{n+1} Z_n = Y_{n-1} + X_n Z_{n-1} + X_{n+1} Z_n$$

$$
\begin{aligned}
\mathbb{E}[Y_{n+1}|X_1,...,X_n] &= \mathbb{E}[Y_{n-1} + X_n Z_{n-1} + X_{n+1} Z_n | X_1,...,X_n] \\
&= \mathbb{E}[Y_{n-1}|X_1,...,X_n] + \mathbb{E}[X_n Z_{n-1}|X_1,...,X_n] + \mathbb{E}[X_{n+1} Z_n|X_1,...,X_n] \\
&= \mathbb{E}[Y_{n-1}|X_1,...,X_{n-1}] + X_n Z_{n-1} + Z_n \mathbb{E}[X_{n+1}] \\
&\quad \text{(since } Y_{n-1} \text{ ind. } X_n \text{, and } X_{n+1} \text{ ind. } X_1,...,X_n) \\
&= \mathbb{E}[\mathbb{E}[Y_n|X_1,...,X_{n-1}]|X_1,...,X_{n-1}] + X_n Z_{n-1} + 0 \quad \text{(induction hypothesis)} \\
&= \mathbb{E}[Y_n|X_1,...,X_{n-1}] + X_n Z_{n-1} \\
&= Y_{n-1} + X_n Z_{n-1} \\
&= Y_n
\end{aligned} \tag{5}
$$

So

$$\forall n \geq 0, \mathbb{E}[Y_{n+1}|X_1,...,X_n] = Y_n$$

Then $Y_0, Y_1, ...$ is a martingale with respect to $X_0, X_1, ...$

Question 5

(a) Clearly, M_n is a function of $X_1, ..., X_n$. Since $q/p > 1, M_n > 0, |M_n| = M_n$. Note that

$$\mathbb{E}[X_i] = p - q$$

$$\begin{aligned} \mathbb{E}[|M_n|] &= \mathbb{E}\left[\left(\frac{q}{p}\right)^{X_1+...+X_n}\right] \\ &= \mathbb{E}\left[\prod_{i=1}^{n}\left(\frac{q}{p}\right)^{X_i}\right] \\ &= \prod_{i=1}^{n}\mathbb{E}\left[\left(\frac{q}{p}\right)^{X_i}\right] \quad \text{(by independence)} \\ &< \infty \end{aligned} \tag{6}$$

$$\begin{aligned} \mathbb{E}[M_{n+1}|X_1, ..., X_n] &= \mathbb{E}\left[\left(\frac{q}{p}\right)^{X_1+...+X_{n+1}} |X_1, ..., X_n\right] \\ &= \mathbb{E}\left[\left(\frac{q}{p}\right)^{X_1+...+X_n}\left(\frac{q}{p}\right)^{X_{n+1}} |X_1, ..., X_n\right] \\ &= \mathbb{E}\left[\left(\frac{q}{p}\right)^{X_{n+1}}\right]\mathbb{E}\left[\left(\frac{q}{p}\right)^{X_1+...+X_n} |X_1, ..., X_n\right] \quad (X_{n+1} \text{ ind. } X_1, ..., X_n) \\ &= \left[p\left(\frac{q}{p}\right) + q\left(\frac{q}{p}\right)\right]\left(\frac{q}{p}\right)^{X_1+...+X_n} \\ &= M_n \end{aligned} \tag{7}$$

So M_i is a martingale with respect to the sequence (X_i).

(b) We have $\mathbb{E}[T] < \infty$.

$$\begin{aligned}
\mathbb{E}[|M_{\min\{i+1,T\}} - M_{\min\{i,T\}}||X_1, ..., X_i] &= \mathbb{E}\left[\left|\left(\frac{q}{p}\right)^{X_1+...+X_{i+1}} - \left(\frac{q}{p}\right)^{X_1+...+X_i}\right||X_1, ..., X_i\right] \\
&= \mathbb{E}\left[\left|\left(\frac{q}{p}\right)^{X_1+...+X_i}\left(1 - \left(\frac{q}{p}\right)^{X_{i+1}}\right)\right||X_1, ..., X_i\right] \\
&\leq \left(\frac{q}{p}\right)^i \frac{q-p}{p}
\end{aligned} \tag{8}$$

By the Optional Stopping Theorem, $\mathbb{E}[M_T] = \mathbb{E}[M_1] = 1$.

Let a be the probability the gamber wins. Then

$$\mathbb{E}[M_T] = \mathbb{E}[(q/p)^{S_i}] = a\left(\frac{q}{p}\right)^n + (1-a)\left(\frac{q}{p}\right)^{-k} = 1$$

We have

$$a = \frac{1 - \left(\frac{p}{q}\right)^k}{\left(\frac{q}{p}\right)^n - \left(\frac{p}{q}\right)^k}$$

(c) Clearly, Y_n is a function of $X_1, ..., X_n$.

$$\mathbb{E}[|Y_n|] \leq i + (q-p)i < \infty$$

Note that

$$\begin{aligned}
Y_{i+1} &= S_{i+1} + (q-p)(i+1) \\
&= S_i + X_{i+1} + (q-p)(i+1) \\
&= Y_i + X_{i+1} + q - p
\end{aligned} \tag{9}$$

$$\begin{aligned}
\mathbb{E}[Y_{n+1}|X_1, ..., X_n] &= \mathbb{E}[Y_n + X_{i+1} + q - p|X_1, ..., X_n] \\
&= Y_n + \mathbb{E}[X_{i+1}] + q - p \quad \text{(independence)} \\
&= Y_n + p - q + q - p \\
&= Y_n
\end{aligned} \tag{10}$$

Thus, Y_i is a martingale with respect to the sequence (X_i).

(d) We have $\mathbb{E}[T] < \infty$.

$$\begin{aligned} \mathbb{E}[|Y_{\min\{i+1,T\}} - Y_{\min\{i,T\}}||X_1, ..., X_i] &= \mathbb{E}[|S_{i+1} + (q-p)(i+1) - S_i + (q-p)i||X_1, ..., X_i] \\ &= \mathbb{E}[|X_{i+1} + q - p||X_1, ..., X_i] \\ &\leq 1 + q - p \end{aligned} \tag{11}$$

By the Optional Stopping Theorem, $\mathbb{E}[Y_T] = \mathbb{E}[Y_1] = 0$.

$$\mathbb{E}[Y_T] = \mathbb{E}[S_T + (q-p)T] = 0$$

By linearity of expectation,

$$\mathbb{E}[S_T] + (q-p)\mathbb{E}[T] = 0$$

$$\mathbb{E}[T] = \frac{\mathbb{E}[S_T]}{p-q}$$

Using part (b),

$$\begin{aligned} \mathbb{E}[T] &= \frac{\mathbb{E}[S_T]}{p-q} \\ &= \frac{an + (1-a)(-k)}{p-q} \\ &= \frac{n + k - n\left(\frac{p}{q}\right)^k - k\left(\frac{q}{p}\right)^n}{(p-q)\left[\left(\frac{q}{p}\right)^n - \left(\frac{p}{q}\right)^k\right]} \end{aligned} \tag{12}$$

Question 6

(a) Let X_i be the indicator random variable that an edge $e_i = (u, v)$ is monochromatic. The probability that the edge is monochromatic is the probability that both vertices are coloured the same so $\Pr(X_i = 1) = 1/2$. Let $X(G) = \sum_i X_i$ be the total number of monochromatic edges in graph G. Note that the number of edges is $|E| = m = dn/2$. By linearity of expectation,

$$\mathbb{E}[X(G)] = \frac{dn}{2} \cdot \frac{1}{2} = \frac{dn}{4}$$

(b) Let $G = (V, E)$ and let $G_i = (V, E_i)$ where $E_i = \{e_1, e_2, ..., e_i\} \subseteq E$. Consider the Doob *edge exposure* martingale

$$Z_i = \mathbb{E}[X(G)|G_1, ..., G_i]$$

Note that $Z_0 = \mathbb{E}[X(G)]$. By exposing a new edge, we cannot change the expected nuumber of monochromatic edges by more than 1 so

$$|Z_{i+1} - Z_i| \leq 1$$

We can apply the Azuma-Hoeffding inequality to get

$$\Pr(|Z_m - Z_0| \geq \lambda\sqrt{m}) \leq 2e^{-\lambda^2/2}$$

With high probability, the number of monochromatic edges is within $O(\sqrt{dn \log n})$ from its expected value.